

Vacuum effects on Lunar/Martian simulant consolidation via enzyme-induced carbonate precipitation

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Abstract

As space exploration advances, the development of construction materials suitable for extraterrestrial environments has become a critical research focus. Biocement, due to its sustainability and potential applications with in-situ resource utilization (ISRU), has emerged as a promising approach for extraterrestrial construction. However, its feasibility under extreme space conditions, particularly in vacuum environments, remains largely unexplored. This study investigates the effects of vacuum conditions on the consolidation of lunar and Martian regolith simulants via enzyme-induced carbonate precipitation and compares the results with those under atmospheric pressure. Experimental results indicate that while vacuum conditions have little effect on precipitation content, they significantly reduce the unconfined compressive strength of the consolidated samples. This reduction is primarily attributed to the rapid volatilization of water and ammonia under vacuum, leading to increased porosity and a looser microstructure, which weakens the mechanical performance. These findings based on the principles of ISRU, offer a new perspective on the potential of biocement in extraterrestrial building applications.

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1. Introduction

With the continued advancement of space exploration, the development of construction materials suitable for extraterrestrial environments has become a crucial area of research. This is not only to meet the future construction needs of lunar or Martian bases but also to reduce the high costs of transporting building materials from Earth to space, along with the associated energy consumption and environmental pollution (Karl et al., 2022; Kokkinakis

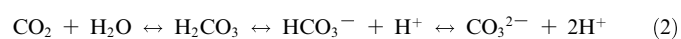
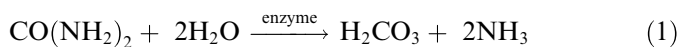
and Drikakis, 2022; Ryan et al., 2022). In response to this challenge, various in-situ resource utilization (ISRU) technologies have been proposed and used for the development of extraterrestrial construction materials, such as sulfur-based concrete (Tute and Goulas, 2024; Giwa et al., 2024; Shahsavari et al., 2022), cast and sintered regolith (Warren et al., 2022; Farries et al., 2021; Indyk and Benaroya, 2017), regolith geopolymer (Javed et al., 2024; Zhang et al., 2022; Montes et al., 2015), and thermoplastic polymers (Fateri et al., 2025; Matetich and Vold, 2023; Zaccardi et al., 2022). Recently, biocement has been researched as a prominent sustainable geotechnical solution for space construction (Zuo et al., 2023; Gleaton

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et al., 2022; Dikshit et al., 2021), with potential advantages such as compatibility with closed-loop resource systems (Yan and Kawasaki, 2025). Biocementation can harness various metabolic pathways, such as ammonification of amino acids, photosynthesis, denitrification, sulfate reduction, and methanogenesis (Castro-Alonso et al., 2019). Among these, urea hydrolysis is the most extensively studied process, as illustrated in Eqs. (1)–(4), in which carbonate ions are generated and subsequently react with calcium ions to form calcium carbonate (Ngoma and Kolawole, 2024; Li et al., 2024; Ahenkorah et al., 2023), generally occurring under neutral to mildly alkaline pH conditions, and thereby enhancing the geotechnical properties of soils in a controlled manner. In particular, enzyme-induced carbonate precipitation (EICP) offers a significant advantage over microbially induced carbonate precipitation (MICP) by eliminating the risks associated with bacterial contamination and not requiring oxygen to sustain activity, which is highly desirable for extraterrestrial applications.

Although the effectiveness and feasibility of biocement technology have been confirmed on Earth, including applications in foundation reinforcement (Shu et al., 2024; Wang et al., 2023; Kou et al., 2023), brick production (Arab et al., 2021; Cheng et al., 2020; Lambert and Randall, 2019), and erosion control (Xu et al., 2024; Naeimi and Chu, 2023; He et al., 2023), the performance of biocement in extreme space environments, which include high vacuum, high radiation, extreme temperature fluctuations, and microgravity, remains largely unexplored. Further research is needed to assess its feasibility and adaptability for extraterrestrial applications, as these conditions could significantly affect reaction kinetics and precipitation characteristics. In particular, this study aims to investigate the feasibility of biocement reactions under vacuum conditions.



The solubility of gases in a solution is proportional to pressure. When environmental pressure decreases, the solubility of gases significantly decreases, leading to an increase in their escape rate (Sander, 2023; Vann et al., 2011; Fernández-Prini et al., 2003). For biocement reactions, the reduction in the solubility of ammonia and carbon dioxide could cause the shifts of Eqs. (2) and (3) to the left, thereby affecting the stability of pH, which is a crucial control factor for the precipitation process. Additionally, the leftward shift in Eq. (2) would reduce the available carbonate ion concentration, potentially altering the rate and amount of calcium carbonate precipitation. Furthermore, the evaporation rate of liquids in a vacuum environment increases significantly, which could lead to dynamic changes in the concentration of reactants in the

solution, thereby impacting the precipitation rate and product characteristics. These factors combined create significant uncertainty in the kinetics and product characteristics of biocement reactions under vacuum conditions. Therefore, to further assess the feasibility and practical application potential of large-scale space biocement technologies, it is crucial to explore the feasibility and specific impacts of biocement reactions under vacuum conditions.

To address these challenges, this study investigates the consolidation of Martian and lunar regolith simulants under simulated vacuum conditions using EICP, as illustrated in Fig. 1, and compares the results with those under atmospheric pressure. By comparing and analyzing the EICP reaction processes and precipitation characteristics under vacuum conditions, the findings provide valuable insights into the behavior of EICP under vacuum, highlighting its potential as a sustainable solution for space applications. This work contributes to the broader understanding of ISRU-based building materials and lays the foundation for applying this technology to space construction and habitat expansion.

2. Materials and methods

2.1. Cementing solution and simulants

In this study, crude soybean urease was selected as the key component of the cementing solution due to its significant cost advantages and practical potential compared to the use of expensive purified commercial ureases (Yan et al., 2025; Liu et al., 2024; Liu et al., 2024). The extraction process of crude soybean urease is as follows: dried soybeans, purchased from the market, were ground and sieved through a 150 μm mesh. The soybean powder was then mixed with distilled water and stirred for 30 min. After filtration, the crude urease solution was obtained without further purification. To balance urease activity and cost efficiency, a concentration of 50 g/L soybean powder was used, resulting in a crude urease solution with an activity of approximately 11 U/mL. This solution was mixed with equal volumes of calcium chloride and urea solutions to form a homogeneous cementing solution. The urease solution and the calcium chloride/urea solutions were only mixed just before use to prevent premature reaction. Both the calcium chloride and urea solutions were at a molar concentration of 1 M, resulting in a final concentration of 0.5 M for both calcium chloride and urea in the cementing solution. Calcium chloride was selected as the calcium source because it is the most commonly used and cost-effective reagent in biocementation research, which facilitates the establishment of a stable and comparable baseline reaction system in preliminary studies.

The simulants used in this study were purchased from Space Resource Technologies. The lunar highland simulant (LHS-1), shown in Fig. 2a, is based on Apollo 16 returned samples and is designed to simulate the characteristics of the lunar surface, specifically the highland regions, which

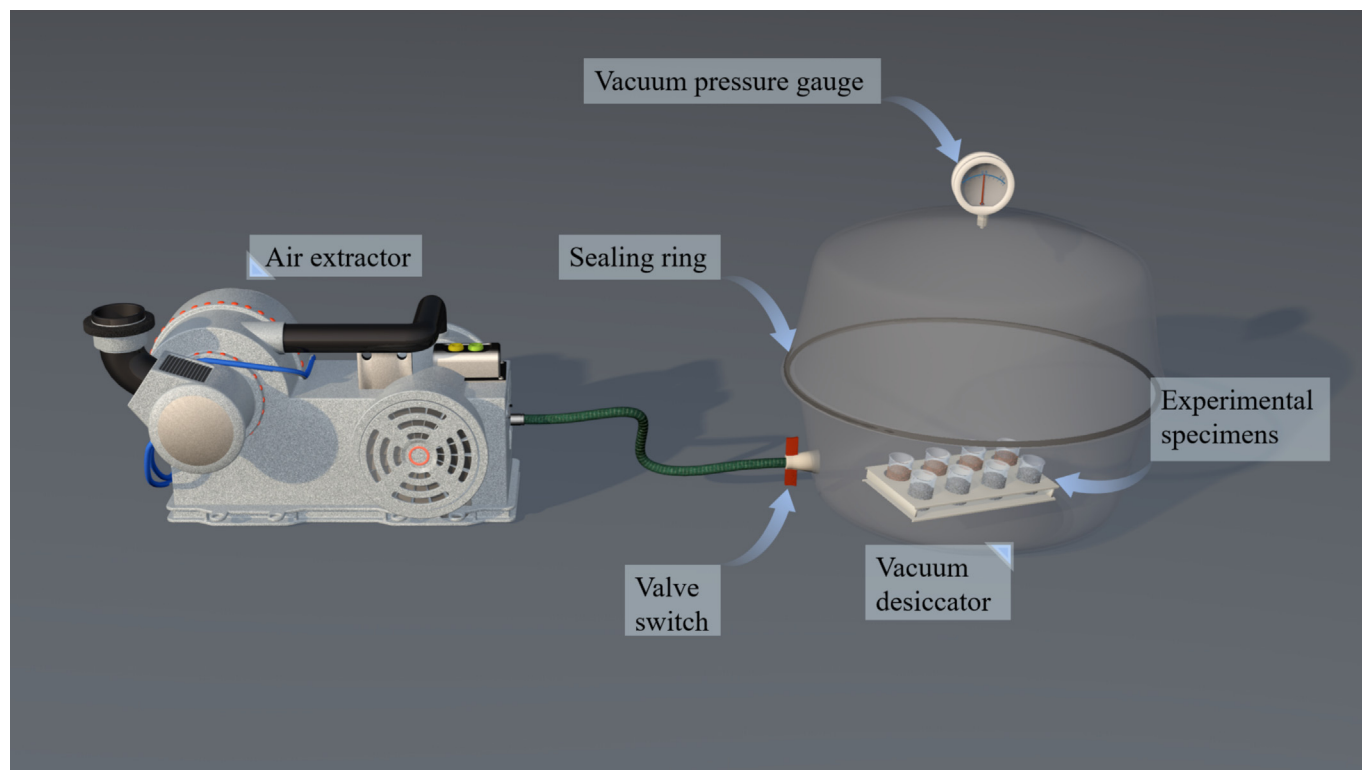


Fig. 1. Schematic diagram of the experimental setup.

are considered the “brighter” areas of the Moon. This simulant is created by precisely combining minerals in proportions that mimic the composition of the lunar regolith. The Martian global simulant (MGS-1), as shown in Fig. 2b, is derived from data collected by the Mars Science Laboratory Curiosity rover and is intended to simulate the Martian surface. This simulant is created by mixing a specific ratio of minerals. Fig. 2c and d present particle size distribution graphs determined via the sieving method, indicating similar particle sizes. Fig. 2e and f display X-ray diffraction (XRD) analysis results obtained using a Rigaku MiniFlex benchtop X-ray diffractometer. Fig. 3a and c show scanning electron microscopy (SEM) images acquired with a Hitachi TM3000 Miniscope, revealing that the simulants consist of angular particles. Fig. 3d and e illustrate zeta potential data obtained using an Otsuka Electronics ELSZ-1000 zeta potential analyzer, showing that both simulants have similar negative zeta potentials. Fig. 3b presents X-ray fluorescence (XRF) data measured with a Hitachi SEA1200VX XRF analyzer, indicating that both simulants contain various components in addition to silica dioxide.

2.2. Consolidation of simulants

To investigate the effects of vacuum conditions on the consolidation of simulants via EICP, experiments were conducted under both vacuum and atmospheric pressure conditions, as shown in Table 1. Prior to the experiments,

the prepared crude urease, calcium chloride, and urea solutions were pre-cooled in a 4 °C refrigerator. The low temperature helps suppress enzyme activity (Xue et al., 2024), thereby delaying the initiation of the reaction as much as possible. This pre-cooling step ensures that the reaction does not begin prematurely during the operations prior to applying the vacuum environment, including mixing the solutions, placing the apparatus, and evacuating the chamber with the vacuum pump. This precaution helps to maximize the assurance that the experiment will proceed under the desired vacuum conditions.

After the solutions were cooled to 4 °C to temporarily suppress enzymatic activity, they were promptly removed from refrigeration, rapidly mixed, and injected into approximately 20 mL of the simulant on a 20 °C benchtop to ensure complete saturation before the temperature rose appreciably. Any excess liquid was then carefully removed, leaving only a 1 mm liquid layer on the surface of the simulant. After 24 h, the remaining liquid in the column was drained to complete one cycle of treatment. Additional treatments were performed as specified in Table 1, for a total of 1, 2, 3, and 4 cycles, with three replicates prepared for each sample. The simulants were placed in an open-top mold with a diameter of 3 cm and maintained in a vertical position using a test tube rack to ensure uniform liquid surface area exposure for all samples, as the evaporation rate is influenced by the liquid surface area. For samples exposed to vacuum conditions, they were placed in a vacuum chamber, as shown in Fig. 1, and evacuated using a

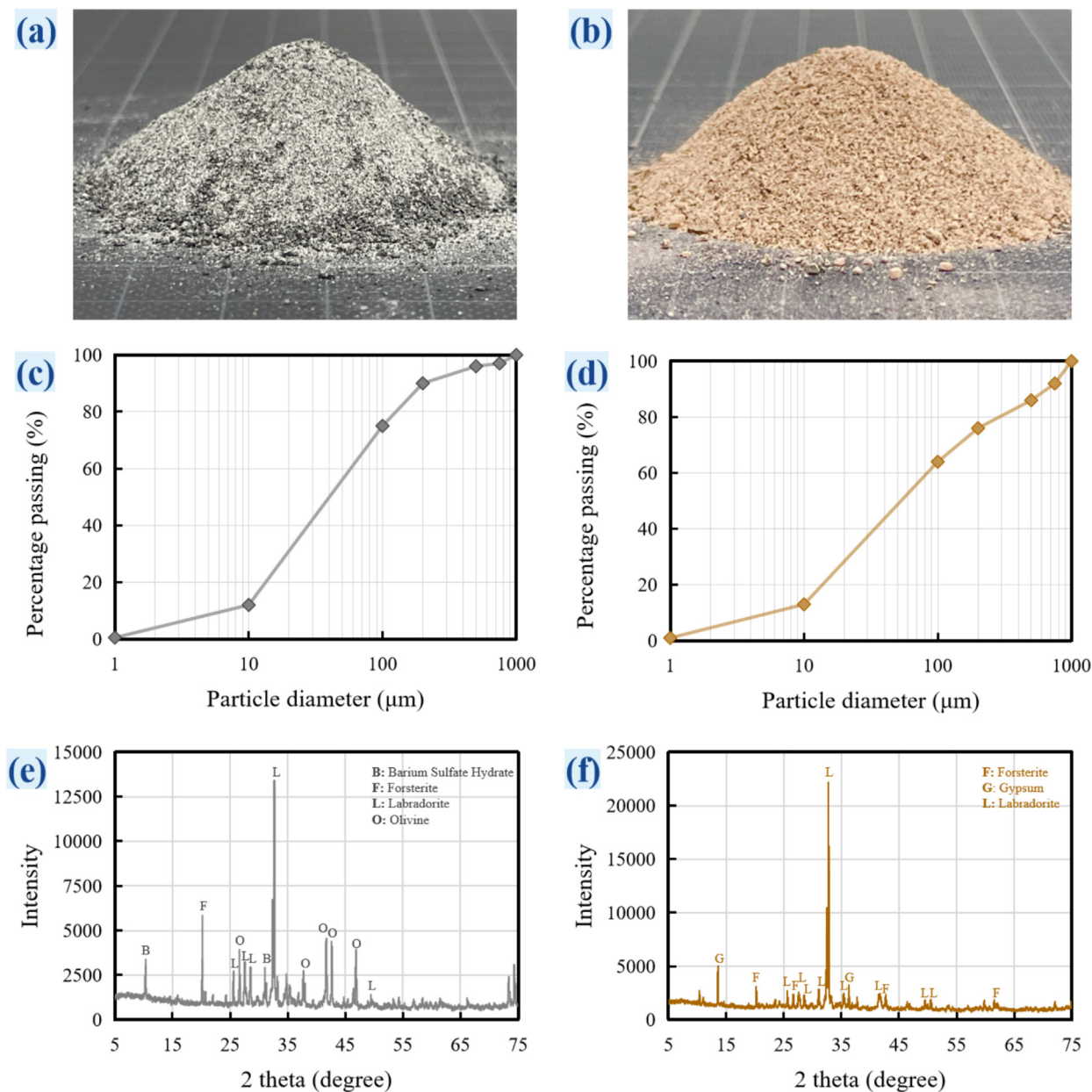


Fig. 2. (a) and (b) show the appearance of the lunar and Martian simulants, respectively. (c) and (d) present their particle size distributions, while (e) and (f) display their crystal phase analyses.

vacuum pump connected to the AS ONE 1-5801-21 vacuum polycarbonate desiccator. However, due to limitations in the vacuum pump's capacity, the minimum achievable pressure was -0.085 MPa (gauge), corresponding to a vacuum level of 83.89%. Samples under atmospheric pressure conditions followed the same procedure and were also maintained at a temperature of 20 °C to ensure that all experimental variables, except for pressure, were consistent.

2.3. Evaluation of consolidation effectiveness

After the reaction, the sample columns were first gently rinsed with distilled water to remove any unreacted soluble

materials while avoiding disturbance to the specimen structure. Subsequently, the samples were placed in an oven at 60 °C and dried for 48 h to prepare them for further analysis.

The dried samples, as illustrated in Fig. 4, were then subjected to unconfined compressive strength (UCS) testing to evaluate the mechanical properties of the consolidated samples. Prior to testing, the samples were lightly trimmed with sandpaper at both ends after demolding to remove minor edge irregularities and to obtain a regular cylindrical geometry with a 1:1 height-to-diameter ratio, and compression was applied at a rate of 1 mm/min on the MTS 816 rock test system. The maximum load at failure was recorded to determine the UCS value.

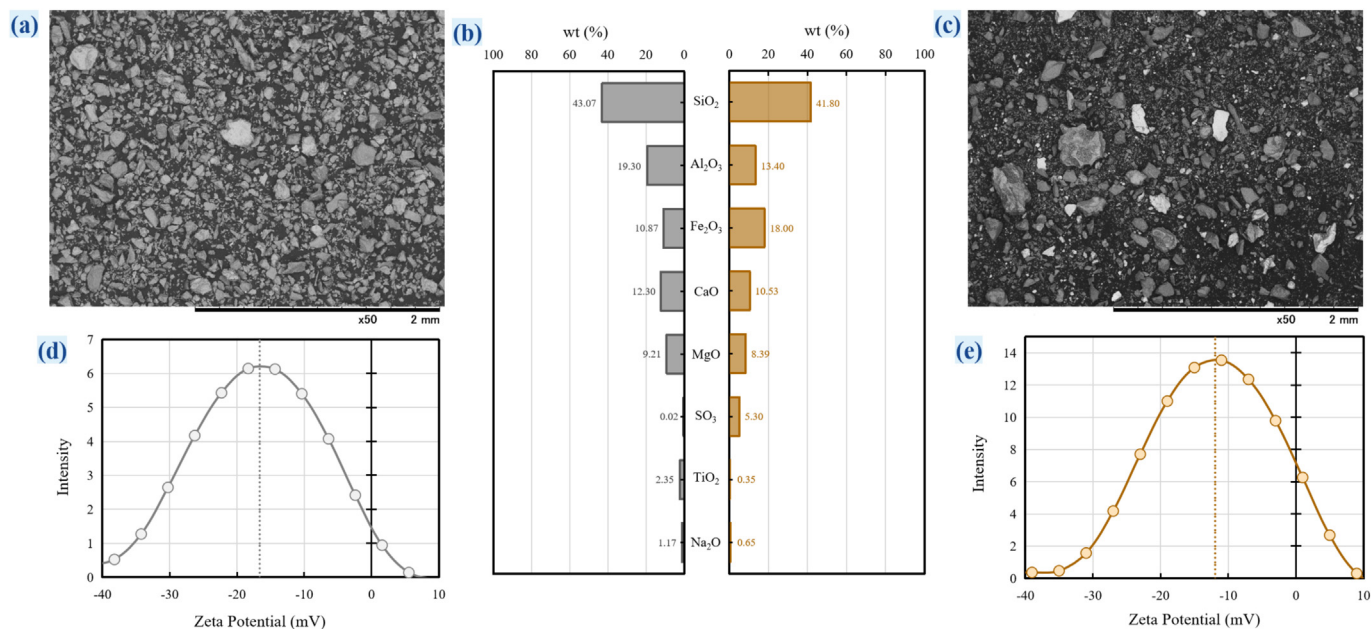


Fig. 3. (a) and (c) show the SEM images of lunar and Martian regolith simulants, respectively. (b) presents their XRF analyses, while (d) and (e) display their zeta potential analyses.

Table 1
Formulations of the samples.

Case name	Pressure condition	Simulant	Treatment time (day)
VL1/VM1	Vacuum	LHS-1/MGS-1	1
VL2/VM2			2
VL3/VM3			3
VL4/VM4			4
AL1/AM1	Atmospheric		1
AL2/AM2			2
AL3/AM3			3
AL4/AM4			4

The calcium carbonate content was determined using the gas pressure method. The dried consolidated samples were ground into a powder to ensure complete reaction. A recorded amount of the powdered sample was placed in a sealed reaction chamber, where 1 M hydrochloric acid was added to dissolve the precipitated calcium carbonate. Prior to the experiments, a pre-calibration test was performed on the lunar/Martian simulants to determine their initial calcium carbonate content. A pressure sensor was used to measure the CO₂ released during the reaction, and based on the CO₂ release together with the initial calcium carbonate content, the amount of calcium carbonate generated during the biocementation process was calculated.

For the analysis of the microstructure and crystalline structure, the microstructure of the samples was observed using a Hitachi TM3000 Miniscope. The central region of each specimen was selected for microstructural observation to avoid capturing only surface features and to ensure comparability across samples. The pore structure and precipitation morphology of the consolidated samples were

recorded. XRD analysis was performed using a Rigaku MiniFlex benchtop X-ray diffractometer with a Co-K α radiation source to analyze the crystalline structure. The scanning rate was set at 6.5°/min, and the scanning range was from 5° to 75°. This analysis aimed to determine whether vacuum conditions affected the crystalline phase of the calcium carbonate.

3. Results and discussion

3.1. Microstructure and crystalline phase analysis

In Fig. 4, it can be observed that under vacuum conditions, the number of pores gradually decreases with increasing treatment cycles. The SEM images in Fig. 5 further confirm this observation and reveal that samples treated under vacuum exhibit larger pores compared to those cured under atmospheric pressure. The formation of these pores may be attributed to the rapid volatilization of water and ammonia, leading to a looser microstructure. Since microstructural characteristics directly influence the macroscopic mechanical properties of materials, this increased porosity is expected to significantly reduce the UCS of the consolidated samples (Long et al., 2023). Furthermore, as the curing time increases and the number of treatment cycles increases, the initially observed voids appear to be gradually filled by continuous precipitation, and over time, the structure becomes more compact. This suggests that increasing the number of treatment cycles may partially alleviate the negative effects of rapid gas escape on the final consolidation quality. Future research could employ techniques such as mercury intrusion porosimetry and micro-computed tomography to quantita-

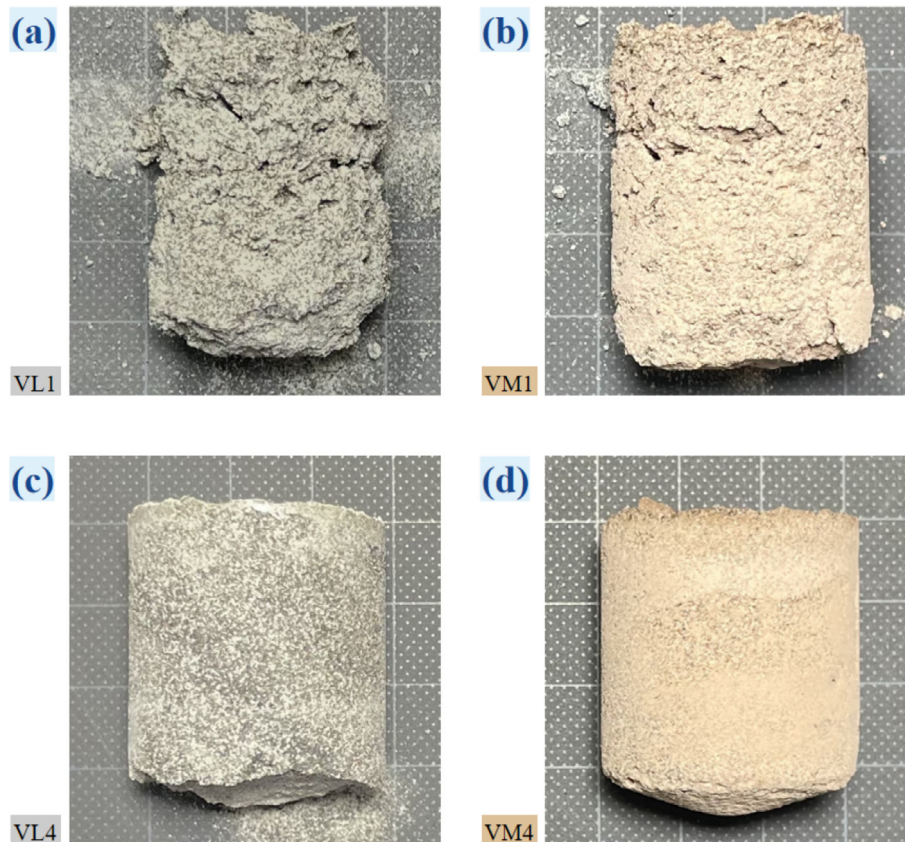


Fig. 4. Appearance of the samples under vacuum and atmospheric pressure conditions, with a 1 × 1 cm grid beneath for scale.

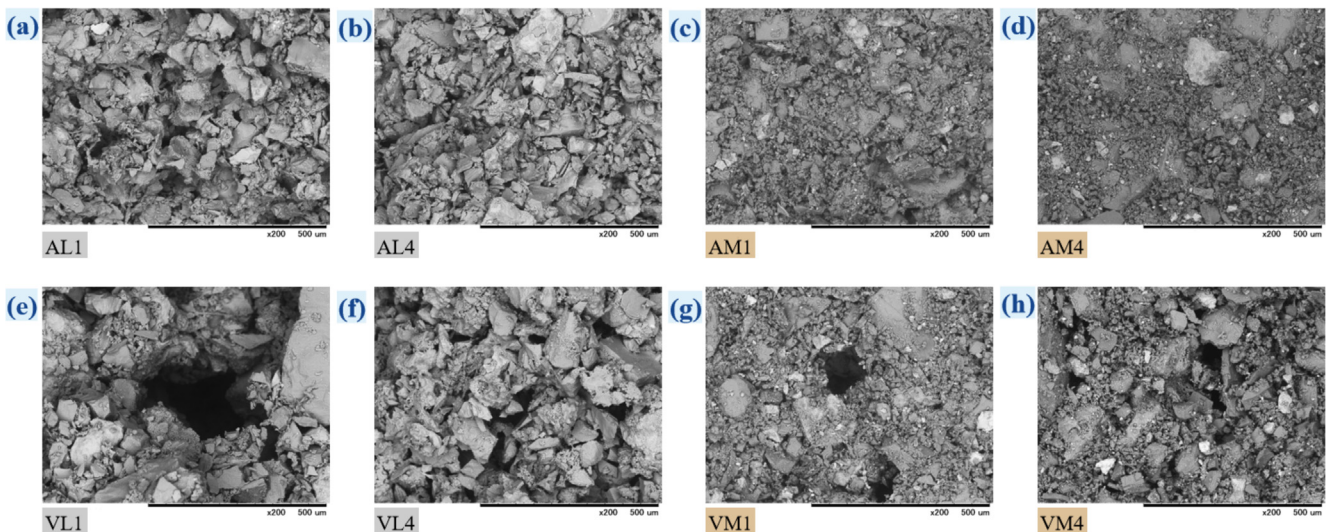


Fig. 5. Microstructure of the specimens after treatment.

tively assess the effects of vacuum on pore size distribution, porosity, and spatial structure.

In addition, XRD analysis in Fig. 6 revealed no significant changes in the crystalline phase composition. This indicates that, although vacuum conditions altered the microstructural characteristics of the biocement material, the mineral composition and crystalline phases of the precipitates remained unchanged (Chang et al., 2022).

3.2. Consolidation performance analysis

Consolidation performance is typically assessed by measuring the precipitation content and the UCS, which are used to evaluate the precipitation efficiency and consolidation effectiveness of biocement. Fig. 7 shows the precipitation content of the consolidated samples under different experimental conditions. The experimental results indicate

that there is no significant difference in precipitation content between different simulants, and the variations in precipitation content under vacuum and atmospheric pressure conditions are relatively small. However, the precipitation under vacuum conditions is slightly lower than that under atmospheric pressure, with the decrease not exceeding 1%. This phenomenon may be attributed to the accelerated evaporation of water in the vacuum environment, which causes the solute concentration in the solution to rise rapidly. However, high concentrations of urea and calcium ions may inhibit enzyme activity (Yan et al., 2024; Xie et al., 2023; Xiao et al., 2023), thereby affecting the efficiency of calcium carbonate precipitation. Additionally, the higher rate of ammonia escape in the vacuum environment may lower the pH of the reaction system compared to the atmospheric pressure environment, further influencing the precipitation kinetics of calcium carbonate and resulting in a slight reduction in precipitation. In future work, monitoring the amounts and release rates of ammonia, carbon dioxide, and water during the reaction process would help to elucidate the precipitation kinetics under vacuum conditions.

Fig. 8 presents the UCS results of the consolidated samples under different experimental conditions. Overall, the UCS in the vacuum environment is significantly lower than

that in the atmospheric pressure environment, indicating that the mechanical properties of the consolidated samples are affected under vacuum conditions. The primary cause of this phenomenon may be the rapid volatilization of water and gases under vacuum, leading to the formation of more pores in the precipitated structure (Lin et al., 2025), which consequently weakens its mechanical strength.

Overall, UCS under vacuum conditions is significantly lower than under atmospheric pressure, while the changes in precipitation content are relatively small. This indicates that, although the EICP process is still feasible under vacuum conditions, its consolidation effectiveness is influenced by changes in the pore structure. Future research should focus on further optimizing the process to improve consolidation strength (Dikshit et al., 2022), for example by introducing suitable additives to increase the viscosity of the solution and reduce liquid evaporation under vacuum conditions, thereby effectively enhancing the performance of EICP in vacuum environments.

4. Conclusions and future prospects

This study represents the first investigation into the feasibility of EICP for consolidating lunar and Martian rego-

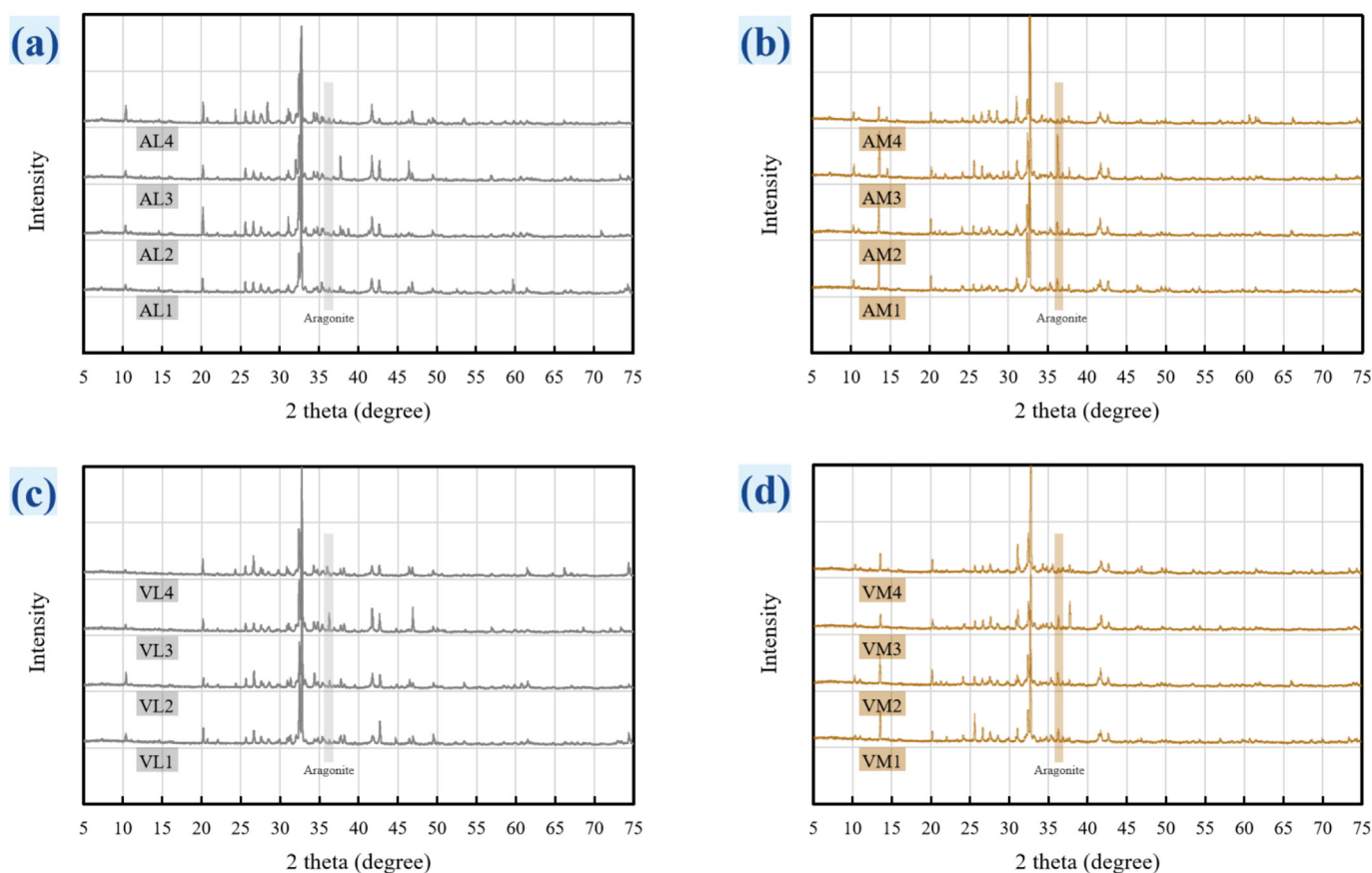


Fig. 6. XRD of the specimens after treatment.

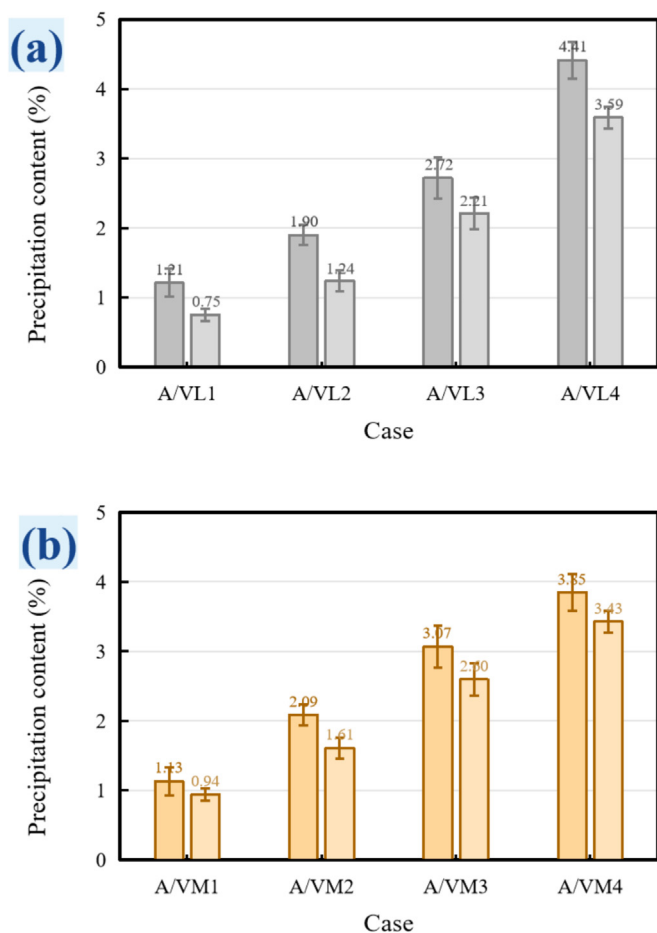


Fig. 7. Precipitation amounts of different samples after treatment.

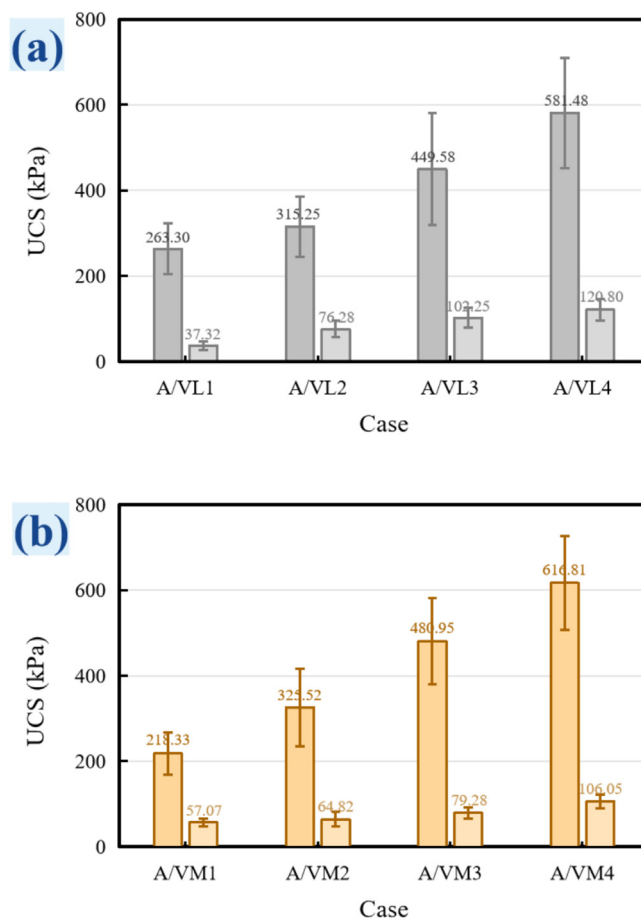


Fig. 8. UCS of different samples after treatment.

lith simulants under simulated vacuum conditions. The results demonstrate that while the vacuum environment has minimal impact on the total calcium carbonate content, it significantly reduces the unconfined compressive strength. This reduction is attributed to the formation of a more porous microstructure caused by the rapid volatilization of water and ammonia under reduced pressure. Scanning electron microscopy analysis confirmed that vacuum-treated samples exhibit markedly higher porosity compared to those cured under atmospheric conditions, directly compromising their mechanical performance. These findings indicate that although EICP remains chemically viable in vacuum, its consolidation effectiveness is limited by structural changes at the microscale. This underscores the need to systematically investigate how vacuum influences porosity and sample density and to optimize reaction parameters such as solution concentration, injection rate, and curing protocol to mitigate pore formation and enhance the strength of biocemented regolith.

Nevertheless, this study has several limitations. The experimental setup achieved only approximately 85% vacuum, which falls short of replicating the extreme high-vacuum conditions present on the Moon or Mars. This discrepancy may lead to an underestimation of ammonia and

water vapor escape rates, potentially affecting the accuracy of predictions regarding reaction kinetics and pH evolution in real extraterrestrial environments. Furthermore, the current work did not account for other critical space environmental factors, including microgravity, cosmic radiation, and extreme temperature fluctuations. For instance, microgravity may suppress convective transport and lead to non-uniform precipitation (Lutz et al., 2023); cosmic radiation may degrade enzyme activity (Rodacka et al., 2016); and extreme temperature fluctuations may impair enzyme stability (Feder et al., 2021). These coupled effects may further influence both the biochemical reaction and the resulting mechanical integrity of the consolidated material.

To advance EICP toward practical ISRU applications, future research should focus on developing experimental systems capable of achieving ultra-high vacuum levels that more closely mimic actual space conditions. It is also essential to conduct multi-factor coupling experiments that integrate vacuum with cosmic radiation, thermal cycles, and microgravity analogs, thereby providing a more realistic assessment of EICP performance. When this technology enters the engineering phase in the future, the impact of each key factor needs to be systematically evaluated in order to construct a complete operational window. Additionally, evaluations using high-fidelity lunar or Martian

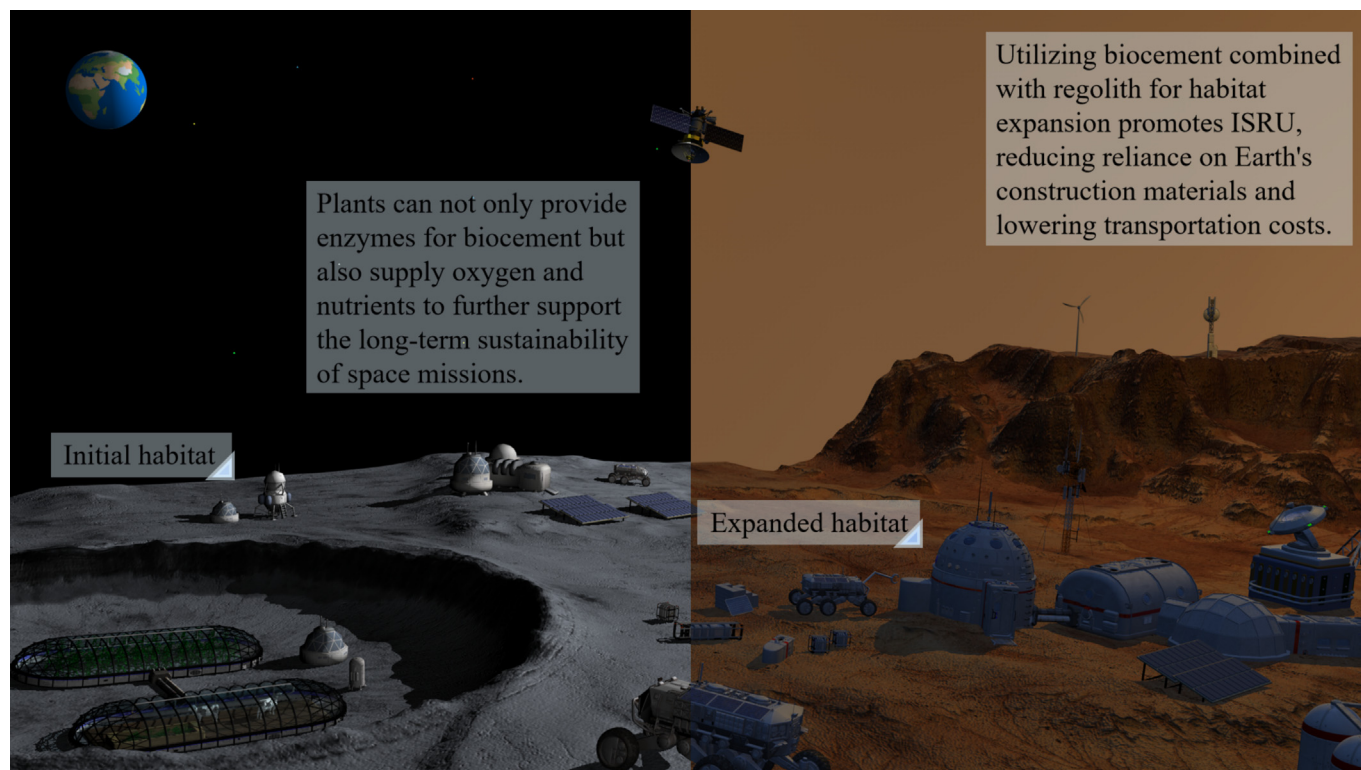


Fig. 9. Schematic of future applications.

regolith simulants as base materials are crucial for understanding the technology's behavior in practical scenarios, particularly considering the potential presence of perchlorates, which are strong oxidizers that may significantly inhibit enzyme activity and alter biocementation outcomes. Exploring synergies between biocementation and existing *in-situ* construction strategies will further support the integration of EICP into long-term, sustainable infrastructure development beyond Earth. For example, EICP has potential applications in the crack repair of structural elements (Jedrzejsko et al., 2025); particularly considering that extraterrestrial structures may experience microcracking caused by extreme thermal cycling or micrometeoroid impacts. As conceptually illustrated in Fig. 9, successfully addressing these challenges could enable EICP to evolve into a robust, Earth-independent technology for habitat construction during long-duration space missions, reducing reliance on interplanetary transport and opening new opportunities for deep space exploration.

Data availability statement

All the experimental data that support the findings of this study are available from the corresponding author upon reasonable request through email.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Ahenkorah, I., Rahman, M.M., Karim, M.R., Beecham, S., 2023. Unconfined compressive strength of MICP and EICP treated sands subjected to cycles of wetting-drying, freezing-thawing and elevated temperature: experimental and EPR modelling. *J. Rock Mech. Geotech. Eng.* 15, 1226–1247. <https://doi.org/10.1016/j.jrmge.2022.08.007>.
- Arab, M.G., Omar, M., Almajed, A., Elbaz, Y., Ahmed, A.H., 2021. Hybrid technique to produce bio-bricks using enzyme-induced carbonate precipitation (EICP) and sodium alginate biopolymer. *Constr. Build. Mater.* 284, 122846. <https://doi.org/10.1016/j.conbuildmat.2021.122846>.
- Castro-Alonso, M.J., Montañez-Hernandez, L.E., Sanchez-Muñoz, M.A., Macías Franco, M.R., Narayanasamy, R., Balagurusamy, N., 2019.

- Microbially induced calcium carbonate precipitation (MICP) and its potential in bioconcrete: microbiological and molecular concepts. *Front. Mater.* 6. <https://doi.org/10.3389/fmats.2019.00126>.
- Chang, H., Wang, X., Wang, Y., Li, S., Wang, J., Liu, J., Feng, P., 2022. Influence of low vacuum condition on mechanical performance and microstructure of hardened cement paste at early age. *Constr. Build. Mater.* 346, 128358. <https://doi.org/10.1016/j.conbuildmat.2022.128358>.
- Cheng, L., Kobayashi, T., Shahin, M.A., 2020. Microbially induced calcite precipitation for production of “bio-bricks” treated at partial saturation condition. *Constr. Build. Mater.* 231, 117095. <https://doi.org/10.1016/j.conbuildmat.2019.117095>.
- Dikshit, R., Dey, A., Gupta, N., Varma, S.C., Venugopal, I., Viswanathan, K., Kumar, A., 2021. Space bricks: from LSS to machinable structures via MICP. *Ceram. Int.* 47, 14892–14898. <https://doi.org/10.1016/j.ceramint.2020.07.309>.
- Dikshit, R., Gupta, N., Dey, A., Viswanathan, K., Kumar, A., 2022. Microbial induced calcite precipitation can consolidate martian and lunar regolith simulants. *PLOS ONE* 17, e0266415. <https://doi.org/10.1371/journal.pone.0266415>.
- Farries, K.W., Visintin, P., Smith, S.T., van Eyk, P., 2021. Sintered or melted regolith for lunar construction: state-of-the-art review and future research directions. *Constr. Build. Mater.* 296, 123627. <https://doi.org/10.1016/j.conbuildmat.2021.123627>.
- Fateri, M., Frick, A., Schuler, C., Schubert, T., Hoffmann, J., van der Walt, J., Salinas, M., 2025. 3D printing of flexible parts out of lunar regolith simulant. *Acta Astronaut.* 229, 779–786. <https://doi.org/10.1016/j.actaastro.2025.02.003>.
- Feder, M.J., Akyel, A., Morasko, V.J., Gerlach, R., Phillips, A.J., 2021. Temperature-dependent inactivation and catalysis rates of plant-based ureases for engineered biomineralization. *Eng. Rep.* 3, e12299. <https://doi.org/10.1002/eng2.12299>.
- Fernández-Prini, R., Alvarez, J.L., Harvey, A.H., 2003. Henry’s constants and vapor–liquid distribution constants for gaseous solutes in H₂O and D₂O at high temperatures. *J. Phys. Chem. Ref. Data* 32, 903–916. <https://doi.org/10.1063/1.1564818>.
- Giwa, I., Dempsey, M., Fiske, M., Kazemian, A., 2024. 3D printed sulfur-regolith concrete performance evaluation for waterless extraterrestrial robotic construction. *Autom. Constr.* 165, 105571. <https://doi.org/10.1016/j.autcon.2024.105571>.
- Gleaton, J., Lai, Z., Xiao, R., Zhang, K., Chen, Q., Zheng, Y., 2022. Optimization of mechanical strength of biocemented Martian regolith simulant soil columns. *Constr. Build. Mater.* 315, 125741. <https://doi.org/10.1016/j.conbuildmat.2021.125741>.
- He, J., Liu, Y., Liu, L., Yan, B., Li, L., Meng, H., Hang, L., Qi, Y., Wu, M., Gao, Y., 2023. Recent development on optimization of biocementation for soil stabilization and wind erosion control. *Biogeotechnics* 1, 100022. <https://doi.org/10.1016/j.bgtech.2023.100022>.
- Indyk, S.J., Benaroya, H., 2017. A structural assessment of unfired sintered lunar regolith simulant. *Acta Astronaut.* 140, 517–536. <https://doi.org/10.1016/j.actaastro.2017.09.018>.
- Javed, U., Ahmed Shaikh, F.U., Rahman, A.K.M.S., 2024. Synthesizing lunar regolith-geopolymer emulating lunar positive temperature regime. *Planet. Space Sci.* 244, 105890. <https://doi.org/10.1016/j.pss.2024.105890>.
- Jedrzejko, M.J., Gan, Y., Chen, X., Jonkers, H.M., Luo, H., 2025. Performance evaluation of EICP with organic/non-organic additives for repairing external cracks in cement-based materials. *Constr. Build. Mater.* 458, 139646. <https://doi.org/10.1016/j.conbuildmat.2024.139646>.
- Karl, D., Cannon, K.M., Gurlo, A., 2022. Review of space resources processing for Mars missions: Martian simulants, regolith bonding concepts and additive manufacturing. *Open Ceram.* 9, 100216. <https://doi.org/10.1016/j.oceram.2021.100216>.
- Kokkinakis, I.W., Drikakis, D., 2022. Atmospheric pollution from rockets. *Phys. Fluids* 34, 056107. <https://doi.org/10.1063/5.0090017>.
- Kou, H., Li, Z., Liu, J., An, Z., 2023. Effect of MICP-recycled shredded coconut coir (RSC) reinforcement on the mechanical behavior of calcareous sand for coastal engineering. *Appl. Ocean Res.* 135, 103564. <https://doi.org/10.1016/j.apor.2023.103564>.
- Lambert, S.E., Randall, D.G., 2019. Manufacturing bio-bricks using microbial induced calcium carbonate precipitation and human urine. *Water Res.* 160, 158–166. <https://doi.org/10.1016/j.watres.2019.05.069>.
- Li, J., Zhu, F., Wu, F., Chen, Y., Richards, J., Li, T., Li, P., Shang, D., Yu, J., Viles, H., Guo, Q., 2024. Impact of soil density on biomineralization using EICP and MICP techniques for earthen sites consolidation. *J. Environ. Manage.* 363, 121410. <https://doi.org/10.1016/j.jenvman.2024.121410>.
- Lin, H., Li, Y., Cheng, G., Liang, S., Han, S., Han, B., 2025. Effects of vacuum exposure on the mechanical properties and microstructures of concrete with and without surface coating. *Constr. Build. Mater.* 476, 141340. <https://doi.org/10.1016/j.conbuildmat.2025.141340>.
- Liu, L., Gao, Y., Meng, H., Pan, Q., Wang, Z., Zhou, Y., Liu, B., Cao, X., 2024. Pore-scale, mechanical, and hydraulic properties of EICP-treated sand using crude legume ureases with different protein contents. *Acta Geotech.* 19, 4747–4763. <https://doi.org/10.1007/s11440-023-02211-w>.
- Liu, Y., Gao, Y., Zhou, Y., Meng, H., Li, C., 2024. Evaluation of enzyme-induced carbonate precipitation using crude soybean urease during soil percolation. *Acta Geotech.* 19, 1571–1580. <https://doi.org/10.1007/s11440-023-02040-x>.
- Long, G., Chen, Y., Tang, Z., Shanguan, M., Gao, C., Xu, S., 2023. Drying shrinkage behavior of cement mortar under low vacuum conditions. *Constr. Build. Mater.* 374, 130933. <https://doi.org/10.1016/j.conbuildmat.2023.130933>.
- Lutz, W.E., Azadmanesh, J., Lovelace, J.J., Kolar, C., Coates, L., Weiss, K.L., Borgstahl, G.E.O., 2023. Perfect crystals: microgravity capillary counterdiffusion crystallization of human manganese superoxide dismutase for neutron crystallography. *Npj Microgravity* 9, 39. <https://doi.org/10.1038/s41526-023-00288-x>.
- Matetich, C., Vold, J., 2023. Regolith-based polymer matrix composites for in situ material creation for long-term extraterrestrial missions. *J. Aerosp. Eng.* 36, 04022103. [https://doi.org/10.1061/\(ASCE\)AS.1943-5525.0001507](https://doi.org/10.1061/(ASCE)AS.1943-5525.0001507).
- Montes, C., Broussard, K., Gongre, M., Simicevic, N., Mejia, J., Tham, J., Allouche, E., Davis, G., 2015. Evaluation of lunar regolith geopolymer binder as a radioactive shielding material for space exploration applications. *Adv. Space Res.* 56, 1212–1221. <https://doi.org/10.1016/j.asr.2015.05.044>.
- Naeimi, M., Chu, J., 2023. Biocementation of sand dike against erosion due to overtopping. *Acta Geotech.* 18, 6745–6757. <https://doi.org/10.1007/s11440-023-01934-0>.
- Ngoma, M.C., Kolawole, O., 2024. Porosity and bedding controls on bio-induced carbonate precipitation and mechanical properties of shale and dolomitic rocks: EICP vs MICP. *Biogeotechnics* 2, 100102. <https://doi.org/10.1016/j.bgtech.2024.100102>.
- Rodacka, A., Gerszon, J., Puchala, M., Bartosz, G., 2016. Radiation-induced inactivation of enzymes – molecular mechanism based on inactivation of dehydrogenases. *Radiat. Phys. Chem.* 128, 112–117. <https://doi.org/10.1016/j.radphyschem.2016.05.022>.
- Ryan, R.G., Marais, E.A., Balhatch, C.J., Eastham, S.D., 2022. Impact of rocket launch and space debris air pollutant emissions on stratospheric ozone and global climate. *Earth’s Future* 10, e2021EF002612. <https://doi.org/10.1029/2021EF002612>.
- Sander, R., 2023. Compilation of Henry’s law constants (version 5.0.0) for water as solvent. *Atmospher. Chem. Phys.* 23, 10901–12440. <https://doi.org/10.5194/acp-23-10901-2023>.
- Shahsavari, M.H., Karbala, M.M., Iranfar, S., Vandeginste, V., 2022. Martian and lunar sulfur concrete mechanical and chemical properties considering regolith ingredients and sublimation. *Constr. Build. Mater.* 350, 128914. <https://doi.org/10.1016/j.conbuildmat.2022.128914>.
- Shu, Y., Song, Y., Fang, H., Wang, D., Lu, W., Huang, Y., Zhao, C., Chen, L., Song, X., 2024. Fiber-reinforced microbially induced carbonate precipitation (MICP) for enhancing soil stability: mecha-

- nisms, effects, and future prospects. *J. Build. Eng.* 94, 109955. <https://doi.org/10.1016/j.jobbe.2024.109955>.
- Tute, R.M., Goulas, A., 2024. Mechanical behaviour of sulphur-based Martian regolith concrete processed under CO₂-rich conditions. *Icarus* 417, 116134. <https://doi.org/10.1016/j.icarus.2024.116134>.
- Vann, R.D., Butler, F.K., Mitchell, S.J., Moon, R.E., 2011. Decompression illness. *Lancet* 377, 153–164. [https://doi.org/10.1016/S0140-6736\(10\)61085-9](https://doi.org/10.1016/S0140-6736(10)61085-9).
- Wang, D.-L., Tang, C.-S., Pan, X.-H., Liu, B., Shi, B., 2023. Coupling effect of fiber reinforcement and MICP stabilization on the tensile behavior of calcareous sand. *Eng. Geol.* 317, 107090. <https://doi.org/10.1016/j.enggeo.2023.107090>.
- Warren, P., Raju, N., Ebrahimi, H., Krsmanovic, M., Raghavan, S., Kapat, J., Ghosh, R., 2022. Effect of sintering temperature on microstructure and mechanical properties of molded Martian and Lunar regolith. *Ceram. Int.* 48, 35825–35833. <https://doi.org/10.1016/j.ceramint.2022.07.329>.
- Xiao, R., Liang, B., Wu, F., Huang, L., Lai, Z., 2023. Biocementation of coral sand under seawater environment and an improved three-stage biogrout approach. *Constr. Build. Mater.* 362, 129758. <https://doi.org/10.1016/j.conbuildmat.2022.129758>.
- Xie, D., Zhang, R., Wang, J., 2023. The influence of environmental factors and precipitation precursors on enzyme-induced carbonate precipitation (EICP) process and its application on modification of recycled concrete aggregates. *J. Clean. Prod.* 395, 136444. <https://doi.org/10.1016/j.jclepro.2023.136444>.
- Xu, J., Li, X., Liu, Y., Li, Z., Wang, S., 2024. Evaluation of wind erosion resistance of EICP solidified desert sand based on response surface methodology. *Constr. Build. Mater.* 447, 138119. <https://doi.org/10.1016/j.conbuildmat.2024.138119>.
- Xue, Y., Arulrajah, A., Chu, J., Horpibulsuk, S., 2024. Soybean urease-based EICP stabilization of washed recycled sands derived from demolition wastes cured at low temperatures. *Constr. Build. Mater.* 434, 136735. <https://doi.org/10.1016/j.conbuildmat.2024.136735>.
- Yan, Z., Kawasaki, S., 2025. Review of in situ resource utilization-based biocementation and regolith consolidation techniques for space applications. *Buildings* 15. <https://doi.org/10.3390/buildings15111815>.
- Yan, Z., Nakashima, K., Takano, C., Kawasaki, S., 2024. Feasibility study of enhancing enzyme-induced carbonate precipitation with eggshell waste for sand solidification. *Biogeotechnics* 2, 100108. <https://doi.org/10.1016/j.bgtech.2024.100108>.
- Yan, Z., Nakashima, K., Takano, C., Kawasaki, S., 2025. Strategies for cost-optimized biocement production: a comprehensive review. *World J. Microbiol. Biotechnol.* 41, 67. <https://doi.org/10.1007/s11274-025-04281-2>.
- Zaccardi, F., Toto, E., Santonicola, M.G., Laurenzi, S., 2022. 3D printing of radiation shielding polyethylene composites filled with Martian regolith simulant using fused filament fabrication. *Acta Astronaut.* 190, 1–13. <https://doi.org/10.1016/j.actaastro.2021.09.040>.
- Zhang, R., Zhou, S., Li, F., 2022. Preparation of geopolymer based on lunar regolith simulant at in-situ lunar temperature and its durability under lunar high and cryogenic temperature. *Constr. Build. Mater.* 318, 126033. <https://doi.org/10.1016/j.conbuildmat.2021.126033>.
- Zuo, H., Ni, S., Xu, M., 2023. An assumption of in situ resource utilization for “bio-bricks” in space exploration. *Front. Mater.* 10. <https://doi.org/10.3389/fmats.2023.1155643>.